

Spotter Freight Rate Prediction Assessment

Machine Learning Engineer Assessment

1. Approach

I treated the 48,000 labeled rows in **data/train-test.csv** as the development set and kept the supplied 12,000-row **data/validation.csv** untouched for final prediction. The main goal was to build a model that could generalize across a later time period rather than relying on a random split.

2. Development Split

Part	Rows	Dates	Use
Training	43,147	2025-01-01 to 2025-09-30	Fit candidate models
Development validation	4,853	2025-10-01 to 2025-10-31	Compare models
Spotter validation	12,000	Supplied separately	Final predictions

The development split was chronological. October was held out from January–September so that the validation period followed the training period. After selecting the model, I retrained it on all 48,000 labeled rows and then predicted the 12,000 supplied validation loads.

3. What I Found in the Data

- **Distance was the dominant numerical signal.** Its correlation with posted rate was about 0.909.
- **Equipment mattered.** Reefer, Flatbed, and Dry Van had noticeably different rate distributions.
- **Rate per mile was right-skewed.** Its median was about 2.15, while the maximum was about 14.13, indicating a small high-rate tail.
- **Rates changed over time.** Monthly mean rate-per-mile was not constant during the January–October development period.
- **Some routes were new at validation time.** 1,461 of the 12,000 supplied validation loads were on routes not seen in the development training data, about 12.18%.
- **Weight was not strongly linearly related to posted rate.** The correlation was about 0.035, but I kept the feature because a tree model can capture nonlinear effects.
- **Weight contained unusual values.** I found both negative and missing weights during the checks. I did not treat the negative values as a reason to discard the rows.

4. Model Experiments

I compared several approaches on the October development validation set. The main improvement came from adding the categorical pickup, delivery, and equipment variables to the HistGradientBoosting pipeline.

Experiment	MAE
Numerical HistGradientBoosting baseline	178.94
Categorical HistGradientBoosting	129.18
Categorical HGB with log target	131.08
XGBoost	151.88
Feature-engineered HGB	131.16
No-route feature-engineered HGB	130.66
Route target encoding HGB	130.48
Route frequency HGB	129.48

Tail-weighted HGB	161.24
-------------------	--------

The categorical HistGradientBoosting setup gave the lowest MAE in this set of experiments at 129.18. The final estimator used **max_iter=300**, **learning_rate=0.05**, and **max_leaf_nodes=31**, with random_state=42.

5. Error Check

Metric	Absolute error
Mean	129.18
Median	46.15
75th percentile	89.66
90th percentile	169.64
95th percentile	262.27
99th percentile	1,785.92
Maximum	13,857.90

Most rows had much smaller errors than the mean suggests, but the distribution had a long upper tail. 257 rows (5.30%) had absolute error above 250, and 66 rows (1.36%) were above 1,000.

6. Final Prediction File

After model selection, I retrained the final pipeline on all 48,000 labeled development rows. The resulting **validation_predictions.csv** has exactly 12,000 rows and the required columns **load_id,predicted_rate**.

Before submission, I checked the file for duplicate IDs, missing predictions, and non-positive predictions. The results were 0 duplicate IDs, 0 missing predictions, and 0 non-positive predictions.

7. Scorer Result and December Scenario

I ran the provided scorer with the final prediction file and the December input file. The scorer confirmed all 12,000 validation predictions and all 31 December predictions, then created **scorer_results/candidate_december.png**.

For the fixed December scenario, pickup, delivery, distance, equipment, and weight remain fixed while the date changes from December 1 through December 31, 2025. The final model produced an approximately constant prediction of **\$870.13** for this scenario.

Figure 1. December 2025 chart produced by the supplied scorer. Fixed inputs: Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb.

8. Reproducibility

File	Role
validation_predictions.csv	Final 12,000-row submission file
notebooks/exploration.ipynb	EDA, validation, and model experiments
score.py	Provided validation/scoring script
scorer_results/candidate_december.png	Required December output
requirements.txt	Python dependencies
readme.md	Setup and run instructions

The provided scorer does not calculate the hidden Spotter model metric. Therefore, the MAE values in this report are development-validation results used for model selection, not a claim about the final Spotter score.

9. Summary

The final workflow uses a time-based development split, categorical and numerical features, several model experiments, and a final fit on all labeled development data. The submission file passed the supplied checks, and the required December chart was generated successfully. The remaining evaluation is Spotter's hidden validation after submission.